Stacking Galaxies Reveals a Hidden Cosmic Fountain

Averaging many galaxy ecosystems exposes a strikingly regular, coherent baryon cycle from disk to virial halo.

arXiv:2609.21374 — Stacked baryon fields in cosmological hydrodynamic simulations

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Background

Galaxies grow by continuously pulling in gas and expelling it through feedback, forming fountain-like cycles. Individual ecosystems are too diverse and dynamic for the underlying pattern to emerge clearly.

> **Observational hints of baryon cycles exist, but no concrete, unified picture links them to structure formation.**

What this paper is selling

The authors stack gas velocity, density, and temperature fields across galaxy ensembles in a cosmological simulation. This suppresses galaxy-to-galaxy scatter and reveals a regular fountain pattern matching observations.

Key results to remember

- Stacking unveils smooth, coherent inflow-outflow cones extending from disks to virial radii.
- A simple parameterized fountain model reproduces the stacked velocity field with small residuals.
- The same fountain geometry persists across stellar mass and redshift, expanding with halo size.

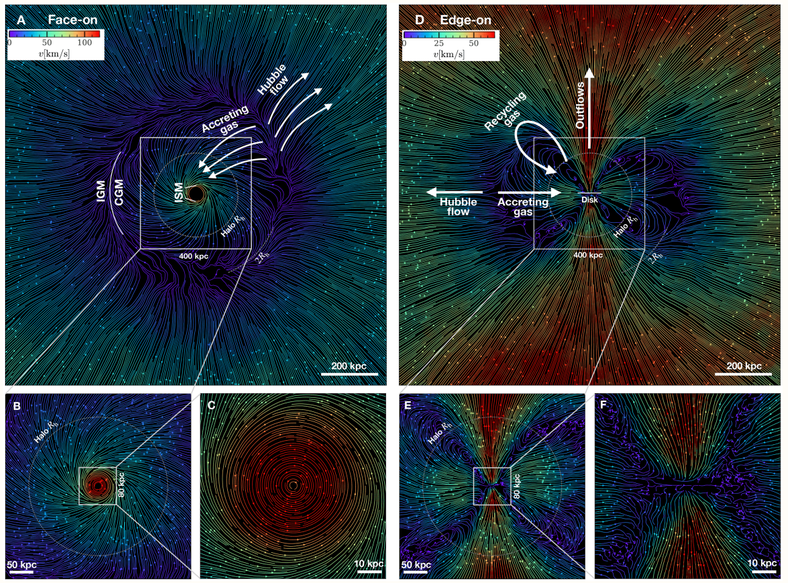


Figure 1 | Stacked velocity streamlines show a smooth, coherent fountain pattern in both face-on and edge-on views.

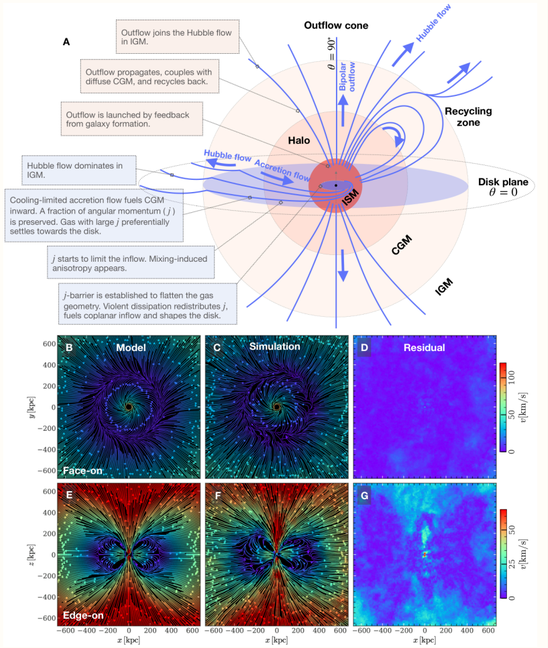


Figure 4 | A simple parameterized fountain model reproduces the simulated stacked velocity field, with residuals shown.